GameboyPi – Raspberry Pi Zero 2 W Handheld im Game-Boy-Color-Format

Strom: LiPo 1S (mit Schutz) -> PowerBoost 1000C (Laden + Load-Sharing + 5.2V Boost) -> +5V

Soft-Power: SW9 EIN haelt PWR_REQ low -> Q1 aus -> EN high. SW9 AUS -> GPIO16 low -> Pi faehrt herunter,

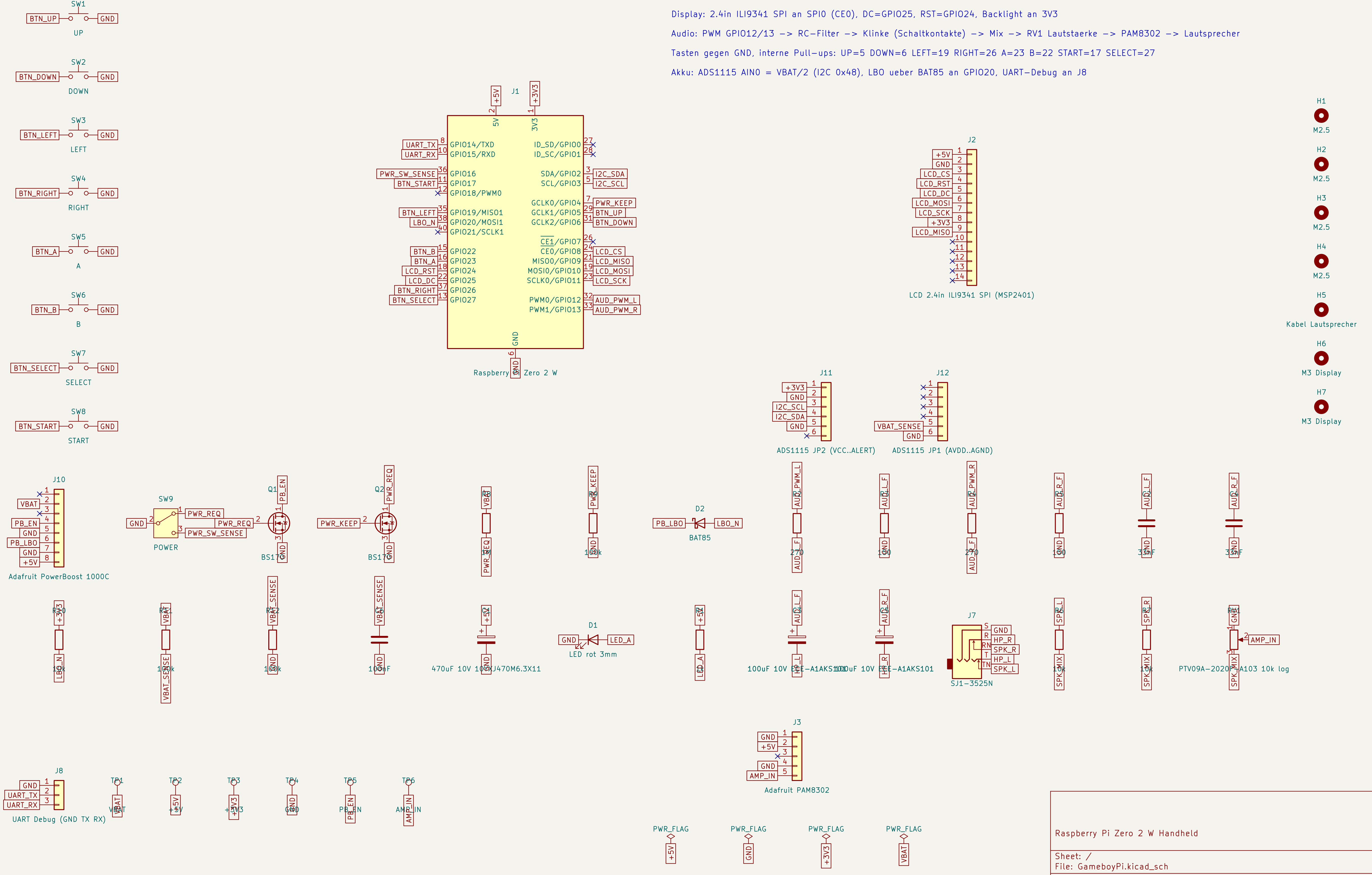
danach GPIO4 (gpio-poweroff) low -> Q2 aus -> R8 zieht PWR_REQ high -> Q1 zieht EN low -> Boost aus.

Display: 2.4in ILI9341 SPI an SPI0 (CE0), DC=GPIO25, RST=GPIO24, Backlight an 3V3

Audio: PWM GPIO12/13 -> RC-Filter -> Klinke (Schaltkontakte) -> Mix -> RV1 Lautstaerke -> PAM8302 -> Lautsprecher

Tasten gegen GND, interne Pull-ups: UP=5 DOWN=6 LEFT=19 RIGHT=26 A=23 B=22 START=17 SELECT=27

Akku: ADS1115 AIN0 = VBAT/2 (I2C 0x48), LBO ueber BAT85 an GPIO20, UART-Debug an J8


SW1 BTN_UP GND UP
SW2 BTN_DOWN GND DOWN
SW3 BTN_LEFT GND LEFT
SW4 BTN_RIGHT GND RIGHT
SW5 BTN_A GND A
SW6 BTN_B GND B
SW7 BTN_SELECT GND SELECT
SW8 BTN_START GND START
J1 Raspberry Pi Zero 2 W
+5V 2 5V
+3V3 1 3V3
UART_TX 8 GPIO14/TXD
UART_RX 10 GPIO15/RXD
PWR_SW_SENSE 36 GPIO16
BTN_START 11 GPIO17
12 GPIO18/PWM0
BTN_LEFT 35 GPIO19/MISO1
LBO_N 38 GPIO20/MOSI1
40 GPIO21/SCLK1
BTN_B 15 GPIO22
BTN_A 16 GPIO23
LCD_RST 18 GPIO24
LCD_DC 22 GPIO25
BTN_RIGHT 37 GPIO26
BTN_SELECT 13 GPIO27
ID_SD/GPIO0 27
ID_SC/GPIO1 28
SDA/GPIO2 3 I2C_SDA
SCL/GPIO3 5 I2C_SCL
GCLK0/GPIO4 7 PWR_KEEP
GCLK1/GPIO5 29 BTN_UP
GCLK2/GPIO6 31 BTN_DOWN
CE1/GPIO7 26
CE0/GPIO8 24 LCD_CS
MISO0/GPIO9 21 LCD_MISO
MOSI0/GPIO10 19 LCD_MOSI
SCLK0/GPIO11 23 LCD_SCK
PWM0/GPIO12 32 AUD_PWM_L
PWM1/GPIO13 33 AUD_PWM_R
GND 6 GND
J2 LCD 2.4in ILI9341 SPI (MSP2401)
1 +5V
2 GND
3 LCD_CS
4 LCD_RST
5 LCD_DC
6 LCD_MOSI
7 LCD_SCK
8 +3V3
9 LCD_MISO
10
11
12
13
14
H1 M2.5
H2 M2.5
H3 M2.5
H4 M2.5
H5 Kabel Lautsprecher
H6 M3 Display
H7 M3 Display
J11 ADS1115 JP2 (VCC..ALERT)
1 +3V3
2 GND
3 I2C_SCL
4 I2C_SDA
5 GND
6
J12 ADS1115 JP1 (AVDD..AGND)
1
2
3
4
5 VBAT_SENSE
6 GND
J10 Adafruit PowerBoost 1000C
1
2 VBAT
3
4 PB_EN
5 GND
6 PB_LBO
7 GND
8 +5V
SW9 POWER
GND 2
1 PWR_REQ
3 PWR_SW_SENSE
Q1 BS170 PB_EN 1 PWR_REQ 2 3 GND
Q2 BS170 PWR_KEEP 2 PWR_REQ 1 3 GND
R8 VBAT PWR_REQ 1M
R9 PWR_KEEP GND 100k
D2 BAT85 PB_LBO LBO_N
R2 AUD_PWM_L AUD_L_F 270
R3 AUD_L_F GND 150
R4 AUD_PWM_R AUD_R_F 270
R5 AUD_R_F GND 150
C2 AUD_L_F GND 33nF
C4 AUD_R_F GND 33nF
R10 +3V3 LBO_N 10k
R11 VBAT VBAT_SENSE 100k
R12 VBAT_SENSE GND 100k
C6 VBAT_SENSE GND 100nF
C1 +5V GND 470uF 10V 10YXJ470M6.3X11
D1 GND LED_A LED rot 3mm
R1 +5V LED_A 1k
C3 AUD_L_F HP_L 100uF 10V EEE-A1AKS101
C5 AUD_R_F HP_R 100uF 10V EEE-A1AKS101
J7 SJ1-3525N
S GND
R HP_R
RN SPK_R
T HP_L
TN SPK_L
R6 SPK_L SPK_MIX 10k
R7 SPK_R SPK_MIX 10k
RV1 GND 1 2 AMP_IN 3 SPK_MIX PTV09A-2020F-A103 10k log
J3 Adafruit PAM8302
1 GND
2 +5V
3
4 GND
5 AMP_IN
J8 UART Debug (GND TX RX)
1 GND
2 UART_TX
3 UART_RX
TP1 VBAT
TP2 +5V
TP3 +3V3
TP4 GND
TP5 PB_EN
TP6 AMP_IN
PWR_FLAG +5V
PWR_FLAG GND
PWR_FLAG +3V3
PWR_FLAG VBAT
1 2 3 4 5 6 7 8
A B C D E F

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